

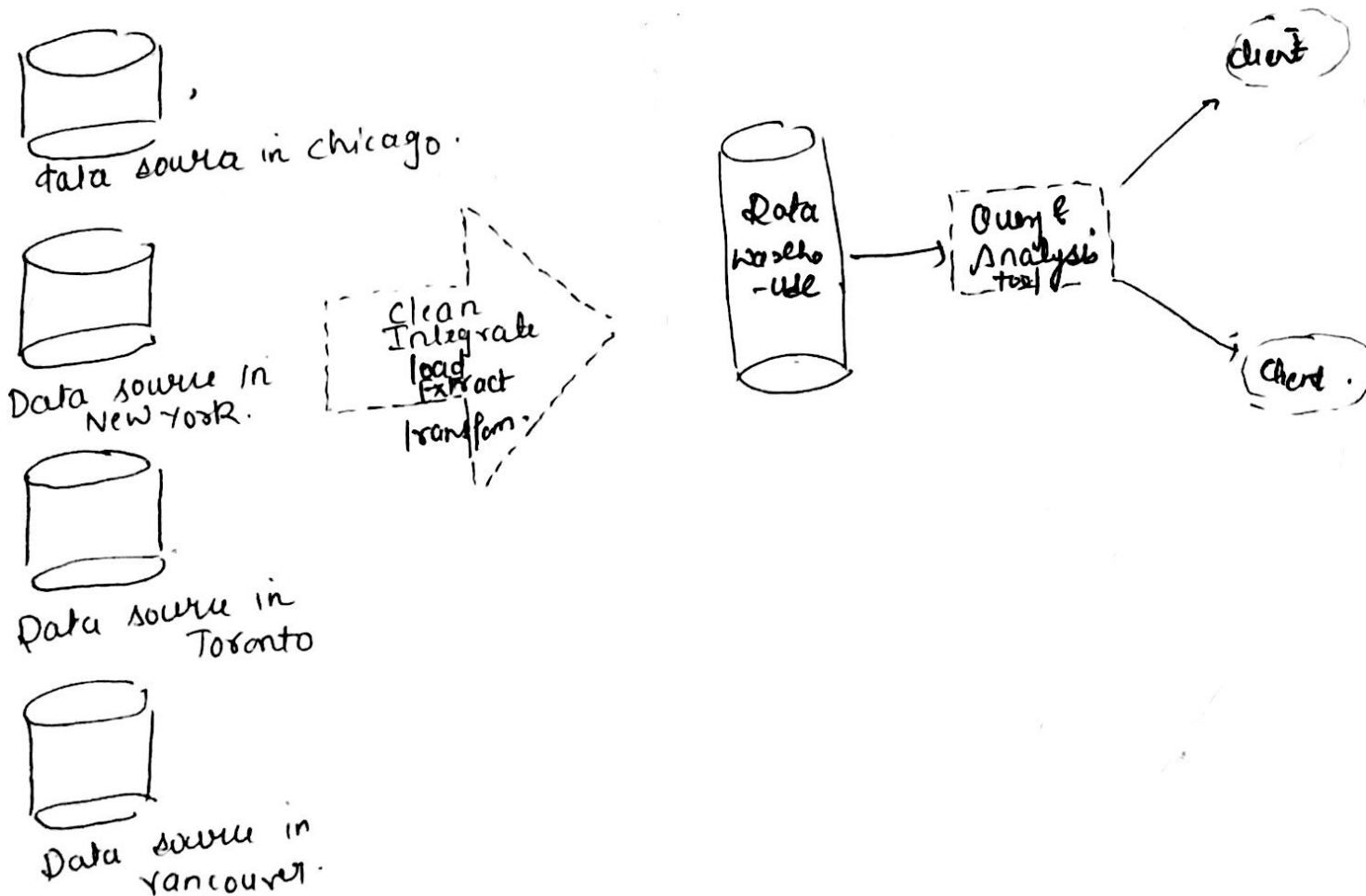
Data Warehouse

Unit-1

Data Warehouse is a repository of Information collected from Multiple sources, stored under a Unified schema & usually residing at a single site.

Data Warehouse is a repository and is constructed via a process of data cleaning, data Integration, data transformation, data loading and periodic data refreshing.

Data Warehouse is usually modelled by Multidimensional data structure called data cube in which each dimension correspond to an attribute or a set of attributes in the schema and each cell store the value.



Features of data warehouse.

③.

Subject oriented:- Data warehouse designed to help you analyse data.

It is organised around subject such as sales, product, customer.

using warehousing, you can answer question like "Who was our best customer for this item last year".

② Integrated:- Data warehousing must put data from different sources into consistent format.

They may resolve such problem as naming conflict, inconsistency.

③ Non volatile: Data once entered into warehouse, data should not be changed.

4) Time variant:- It provides analyst need large amount of data. A data warehouse has a change over time that is mean by time time variant.

Data cube

A data warehouse is usually modelled by Multidimensional data structure called data cube.

In which each dimension corresponds to attribute.

Data Warehouse Process:

Backend tools & utilities to populate & refresh data. Data warehouse use ④.

① data extraction:- It is the process of ^{gathering} extracting data for warehouse from various sources. data may come from variety of sources

- ① Production data
- ② Metadata
- ③ Legacy data
- ④ Off-line systems

② Data cleaning:- It delete errors in data & rectify them when possible. The techniques include.

- ① Using transformation rules.
- ② Using domain specific knowledge
- ③ Performing parsing & fuzzy matching.

③ Transformation. It convert data from legacy's host format to warehouse format.

④ loading:- It load data by sorting, summarizing, verifying, checking integrity & partitioning

- ① Sequential loading
- ② Incremental loading
- ③ Batch loading

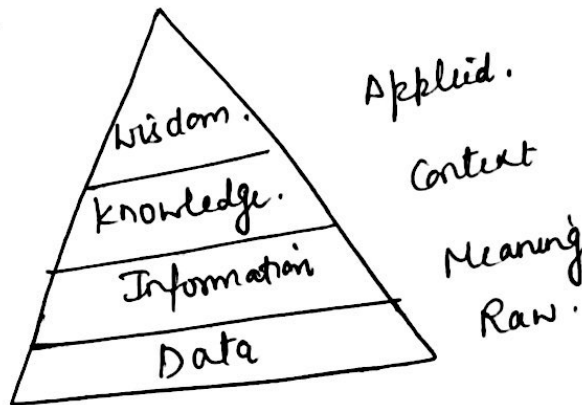
⑤ Refresh:- It propagate the update from data source to the warehouse.

Data Mining.

① 1.1
Mining is defined as process of extracting information or patterns from huge set of data in large database.

OR

It is the search for relationship and global pattern that exist in large database but hidden among the vast amount of data.



Data: It is numerical quantity derived from observation, experiment calculation, .

Information: Information is data extracted, filtered or formatted in some way.

Knowledge: It is subset of information. It is subset that has been extracted, filtered or formatted in very special way.

Wisdom: It is the application of knowledge expressed in principle to arrive at sensible & clever decision.

Knowledge is used to guide the search or evaluate the resulting patterns.

Information which are collected

- Mail & Personal Data:- from Government census & personal & customer file, large collection of information
- 2) Satellite sensing → Many satellite pictures & data are made public as soon as they are received
- 3) Business Transaction:- transaction are released to purchase exchange, banking stock.
- 4) Scientific data → data collected from nuclear laboratories, data about ocean activities. Data about human psychology.
- 5) Games:- tremendous amount of data and statistics about games, players, athletes.
→ use this information for Reporting
- 6) World Wide Web Repositories:- All sort of formats, contents & description has been collected.

Motivation Behind Data Mining

The motivation behind Mining data, → the need to find useful information in data to enable better decision making or better understanding of the world.

→ Due to wide availability of data stored in files, database & other Repositories it is necessary to develop powerful means for analysis & interpretation of such data. and the extraction of interesting knowledge that could help in decision making.

Application of Data Mining OR Benefits of data Mining ③. 1.3

Market Analysis & Mgt.

- (1) Customer Profiling
- (2) Identify customer Requirement.
- (3) Determine customer purchasing pattern
- (4) Cross Market Analysis → association b/w product sales.
- (5) Branding Summary Information → Multidimensional summary Report.

②. Fraud Detection

- (1) Credit card services
- (2) Telecommunications → in fraud telephone calls.

(3) Corporate Analysis & Risk Management.

- Finance Planning And Asset Evaluation
- Resource Planning
- Competition → Monitoring competitors & Market direction

Applications

① Sales / Marketing

2) Banking

(3) Insurance And Health care.

(4) Medicine

kind of Data can be mined.

Flat files: These are simple data files in text or binary format. Data in these files can be transactional, scientific measurement, time series data. (4) 1.4

2) Relational Database: - Relational database consist of set of table. Data Mining algorithm use relational database.

3) Data warehouse: - Data warehouse is a storehouse is a repository of data collected from multiple data source. To facilitate decision making and multidimensional view data warehouse are modeled by multidimensional data structure.

4) Transaction Database: - Transaction database is set of records representing transaction with a time stamp.

5) Multimedia Database: - Multimedia database include video, image, audio and text media. Data Mining from Multimedia Repositories may require Computer vision, computer graphics, image interpretation.

6) Spatial Database: - Spatial database store geographical information like maps.

7) World wide web: - Data Mining on web is called web Mining. It consist of three components - content of web which encompass document.

KDDKnowledge discovery in Database.

KDD is the overall process of extracting knowledge from data. Data Mining is the step inside KDD process which deal with identifying pattern in data.

Steps Involved in Knowledge discovery Process.

- (1) Data cleaning
- (2) Data Integration
- (3) Data selection
- (4) Data Transformation
- (5) Data Mining
- (6) Pattern Evaluation
- (7) Knowledge Presentation.

Data cleaning: In this step noise or irrelevant data or inconsistent data is removed. It is data preprocessing technique such as people making typing errors, or clients move from one place to another without notify change of address.

② Data Integration: In this step multiple data sources are combined.

3) Data selection: In this process those data relevant to analysis task are retrieved from database.

4) Data Transformation: The data even after cleaning are not ready for mining, we need to transform them into forms appropriate for Mining.

process of changing the form or structure of attribute. It involve normalisation, aggregation etc. ⑤ i.e

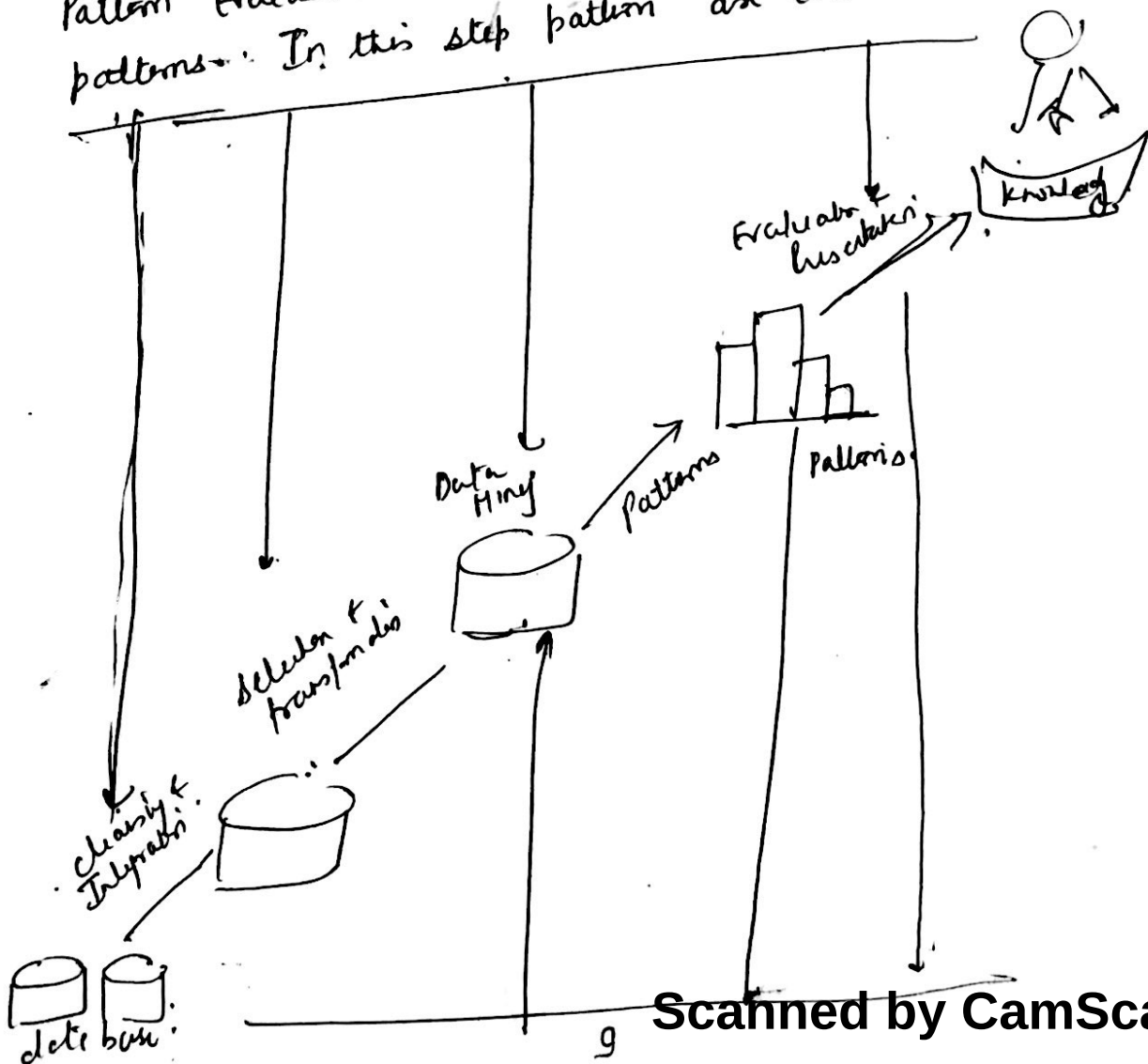
Data Mining: Data Mining techniques on the data to discover interesting pattern. So data mining is the process of discovering interesting knowledge from large amount of data stored either in database, data warehouse.

Techniques like clustering, Association are used for mining.

b) Pattern Evaluation And Knowledge Presentation..

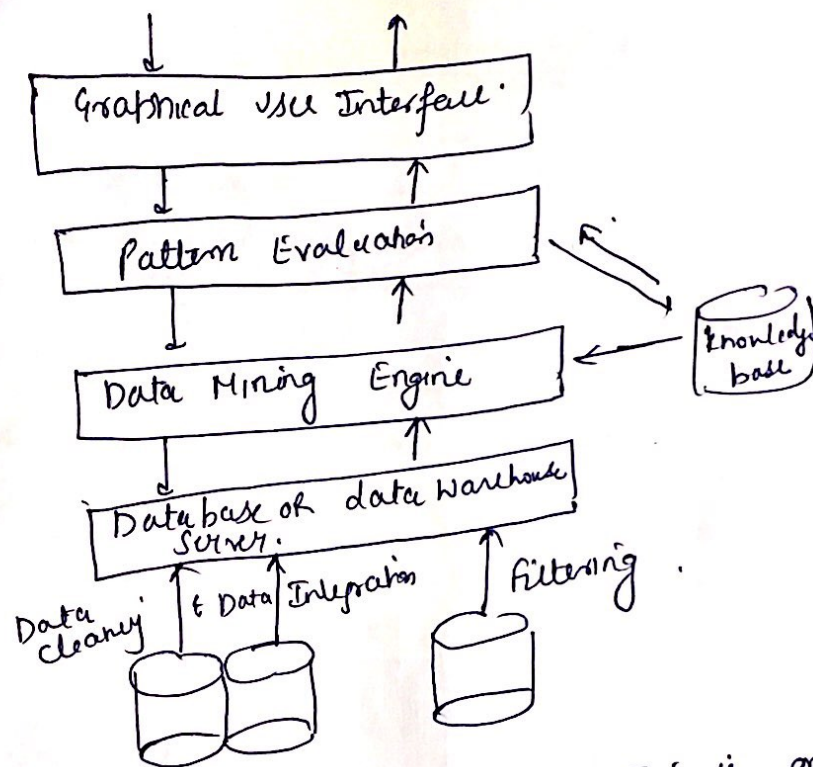
This step deal with how data Mining Result are presented to user. Visualisation technique include graphical & geometrical techniques.

Pattern evaluation is used to identify Truly Interesting patterns. In this step pattern are evaluated.



Architecture of Typical Data Mining system,

⑦ 1.7



- 1) Database, data Warehouse Repository: This is one or set of database, data warehouse, spread sheets or other kind of Information Repository. Data cleaning & data Integration technique may be performed on data.
- 2) Database or data warehouse server: - Database or data warehouse server is responsible for fetching relevant data based on user's data mining request.
- 3) Knowledge Base: - This is the domain knowledge that is used to guide the search or evaluate the resulting patterns.
- 4) Data Mining Engine: - It ideally consist of set of functional modules for task such as association analysis, classification evolution.

Evaluation Module:- In this pattern evaluation module is integrated with data mining module so as to focus the search toward interesting patterns. (Correlation).

6) Graphical User Interface:- This module communicates b/w users & data mining system, allow the user to interact with system by specifying data mining query or task that provide information to help focus the search & perform exploratory data mining.

Data Mining Tasks

It is divided into two categories
Descriptive data Mining:- It provides information to understand what is happening inside the data without a predetermined idea. It characterize general properties of data in database.

Predictive data Mining:- It allows the user to submit records with unknown field values and system will guess unknown value based on previous patterns discovered from database.
Predict value of specific (target / dependant variable) based on value of attribute.

Ex- Judge if a patient has specific disease based on his/her medical test result.
In this mining perform inference on current data in order to make prediction.

Data Mining functionality.

① 1.1

Class / Concept Descriptions : Characterization and Discrimination.

Data can be associated with class or concepts. Such description of class or concept are called class/concept descriptions.

① Data characterization :- This refers to the summarization of general characteristic or features of target class of data. target class (which is under study). For eg:- to study study characteristic of s/w product whose sales increased by 10% last year, one can collect the data related to such products by executing ~~sum~~ SQL query.

② Data discrimination :- It is the comparison of general features of target class data objects with the general features of objects from one or set of contrasting class. Output of data characterization can be represented in various form. Example → include pie charts, bar charts, curves.

mining frequent patterns, Association and Correlations: ②

requent patterns are the patterns that occur frequently in data.

- ① frequent item set: It refers to set of item that frequently appears together for eg:- milk & bread.
- ② frequent subsequence: sequence of pattern that occurs frequently such as purchasing a phone followed by memory card.
- ③ frequent sub structure: substructure refers to different forms (graphs, tree) that may be connected with item-set or subsequence.

Mining Association Analysis:-

uncovering relationships among data & determining Association rules. It is method of discovering relationship b/w different set of items.

(Ex:) Association are used in retail sales to identify pattern that are frequently purchased together.

For eg. A Retailer generate an association rule that show 70% time milk is sold with bread and only 30% of time biscuits are sold with bread.

Association Rules are in the form of $X \Rightarrow Y$. Database tuples that satisfy condition in X are also likely to satisfy condition in Y.

{teabags, sugar} $\Rightarrow$ {milk, snack}.

Ques: Define:- It is the process of finding a model (or function) describe and distinguish data classes or concepts. (3) 1.11

derived Model can be presented in following form.

(1) Classifier (If Then) Rules

(2) Decision trees

(3) Neural network.

(4) Mathematical formulas.

classification can be used for predicting the class label of data objects.

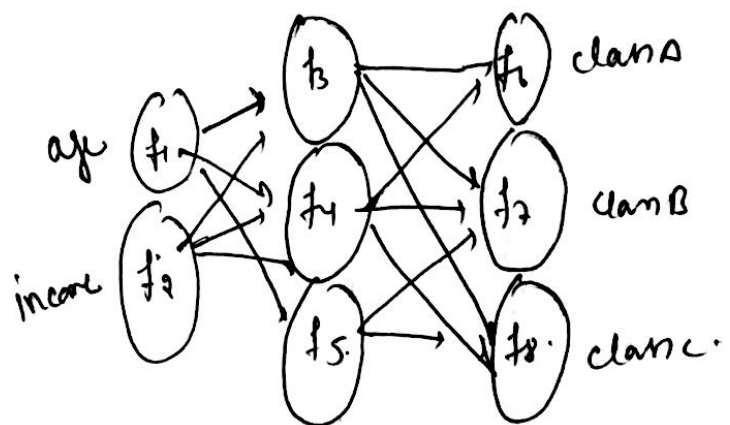
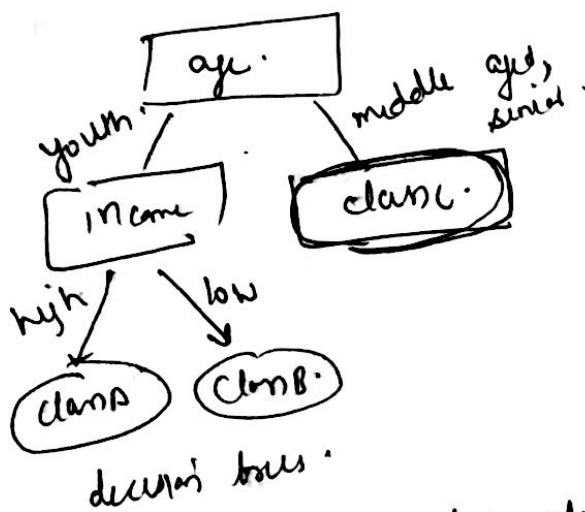
age (x, "young") AND income (x, "high") $\rightarrow$ class (x, "A").

age (x, "young") AND income (x, "low") $\rightarrow$ class (x, "B").

age (x, "middle-aged") $\rightarrow$ class (x, "C");

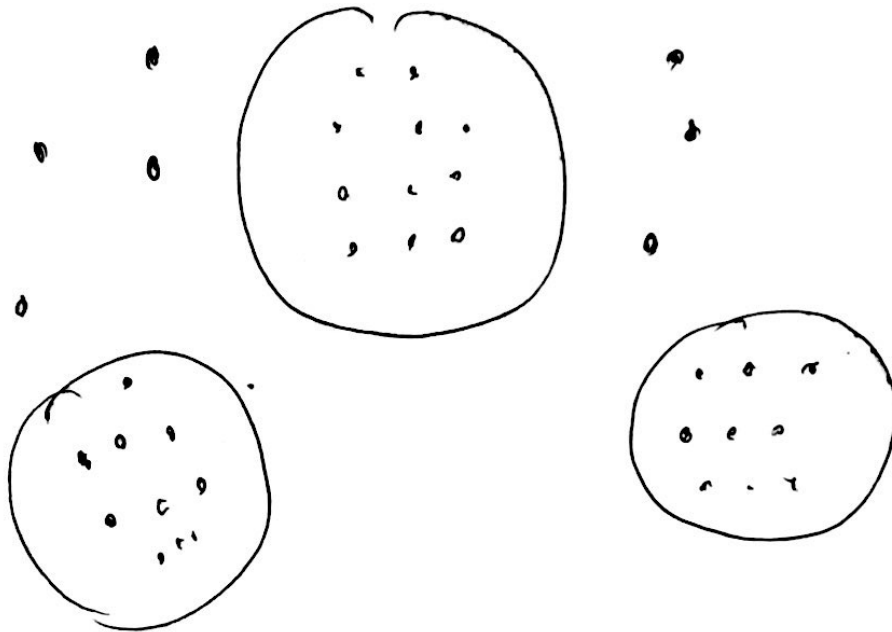
age (x, "senior") $\rightarrow$ class (x, "D").

If then else Rules.



Neural Network used for classification, is typically a collection of neuron like processing units with weighted connections. The weights on connection encode knowledge of network.

Analysis:- clustering analyze data objects without consulting a known class label. clustering can be used to generate such labels. The objects are clustered or grouped based on principle of maximizing intra class similarity & minimize interclass similarity.



Neural network: They can be used to model complex relationship between inputs & outputs & to find pattern in data.

one of the most common application of NN is in image processing. for eg: identify hand written characters; matching a photograph of a person's face with different photo in database. other application voice recognition.

→ Neural network is defined as information processing system. inspired by biological nervous system such as brain. It is composed of highly connected elements (neuron) working in parallel to solve particular Pbm.

→ One of the most important problem in data mining is to determine predictors or raw input data are most important in building model that are most accurate in predictions

Data Mining Issues

Security and Social Issue → Large amount of sensitive & private information about individual or company is gathered or stored. So security is an important issue. Another issue that arises is the appropriate use of data mining.

2) Mining Methodology issue →

Diversity of data available.
Dimensionality of domains
Versatility of mining approach.
Broad analysis need.
Control & Handling of noise in data

3) Data sources issue →

we are storing data in variety of repository. It is difficult to expect a data mining system to effectively & efficiently achieve good mining result.

4) Performance issue →

Many artificial intelligence and statistical method exist for data analysis & interpretation. These method are not designed for very large data set. This raise issue of scalability & efficiency.

5) User Interface Issue →

Good data visualisation eases the interpretation of data mining result as well help user to better understand need. still much research are needed to accomplish in order to obtain good visualisation tool for large dataset that could display used to display & manipulate knowledge.

Mining Methodology.

(i) Mining different kind of knowledge in database: - Different users may be interested in different kind of knowledge. It is necessary for data Mining to cover broad range of knowledge discovery task.

(2) Handling noisy or incomplete data: - The data cleaning methods are required to handle the noise and incomplete objects while mining the data regularities. If the data clean methods are not there then the accuracy of discovered pattern will be poor.

3) Pattern Evaluation: - The pattern discovered should be interesting because either they represent common knowledge or lack novelty.

(2) User Interaction: -

(i) Interactive Mining of knowledge at Multiple level of Abstraction: - Data Mining process need to be interactive because it allow users to focus the search for patterns, providing and refining data Mining request based on returned result.

(ii) Incorporation of Background Knowledge: - To guide discovery process and to express discovered patterns, background knowledge may be used to express discovered pattern.

And visualization of data Mining result. Once patterns are discovered it needs to be expressed in high level language and visual representation. These representation should be easily understandable.

3) Performance Issue:

(i). Efficiency and Scalability of data Mining Algorithm:- In order to extract information from huge amount of data in database, data Mining Algorithm must be efficient and scalable.

(ii). Parallel, distributed and Incremental Mining Algorithm: The factors such as huge size of database, wide distribution of data and complexity of data Mining methods motivate development of parallel & distributed data mining Algorithm.

4) Diverse Data Type Issues:

(i) Handling of Relational and Complex types of data:- The database may contain complex data objects, multimedia data objects, spatial data, temporal data. It is not possible for one system to mine all these kind of data.

(ii) Mining Information from heterogeneous database & Global Information system:

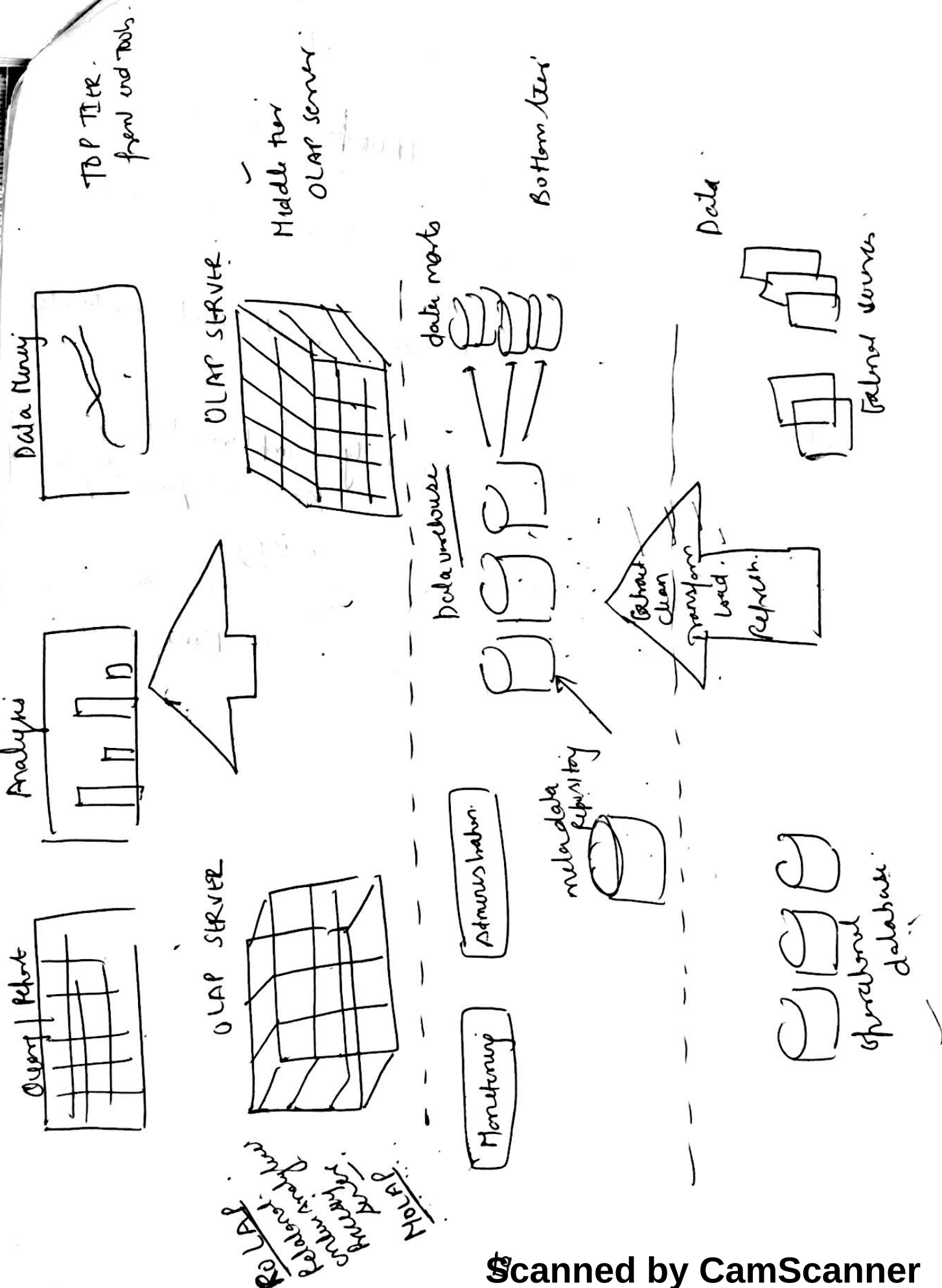
The data is available at different data source on LAN or WAN. These data source may be structured, semi structured or unstructured.

Structured data: The data which can be stored in table with rows & column.

Unstructured data: It includes, mail, msg, video, photo, files, presentation.

Semi structured data: It does not reside in relational database but have some organisational format.

Data Warehousing :- A Multilayer Architecture



Unit-2

Q.1 (1) Data: It is collection of data objects and their attributes.

- An attribute is a property or characteristics of an object.
for eg:- eye color of a person, temperature etc.
attribute is also known as variable, field, characteristic or feature.

- A collection of attributes describe an object. object is also known as entity or instance and Record.

objects ↑

Roll No	Student Name	School Name	Class
1	Rahul	D.A.V. Public School	X
2	Aditya	"	XI
3	Reena	"	XII

[Data set are made up of data object.]

- Data object represent an entity
- Data object typically described by attributes

In sales database, object may be customer and attributes are customer Id, name & address.

Different type of attributes or data types.

- Nominal attributes: nominal attributes provide only enough information to distinguish one object from other.
for eg: Employee Id number, sex & male, female?
male = 0 & female = 1 [Each attribute represent some category].
- ordinal attributes: The value of an ordinal attribute provide enough information to order objects. (1 to 5).
for eg: Rankings
Grades.
height (tall, medium, short).

[ordinal attribute is an attribute with possible values]

- Binary attributes → These are in the form of 0 & 1. These are nominal attribute with only two values true & false.

Interval: For interval attributes, the difference between values are meaningful. These attributes are measured on equal size units. (2:3)
For eg.: calendar dates, temperature in Celsius or Fahrenheit.

1) Ratio: For Ratio variable, both differences and Ratios are meaningful. For eg.: age, Mass, length, electrical current.

For ex: salary, weight.
[In this we can compute difference between values as well as mean, median and mode].

Data Quality:

Why Preprocess the Data.

There are many factors that

compromise data quality.
[Inaccurate, Incomplete and Inconsistent data are common properties of large Real world database].

- 1) Accuracy.
- 2) Completeness
- 3) Consistency
- 4) Timeliness.
- 5) Believability
- 6) Interpretability.

① Accuracy: There are many possible Reason for inaccurate data. (i.e. having incorrect attribute values, These may be human or computer errors.

2) Completeness: Incomplete data can occur for number of reasons. Attributes of interest may not always be available such as customer information for sales

ation data.

Consistency :- Incorrect data may also result from inconsistencies in naming convention or data codes, or inconsistent format for input field. Duplicate steps also require data cleaning.

4) Timeliness :- It also effect data quality for eg:- several sales representative fail to submit their sales record on time at the end of month. There are also a number of corrections & adjustments that flow in after the month's end. For a period of time following each month, data stored in database are incomplete.

(5) Believability :- It reflects how much the data are trusted by users.

(6) Interpretability :- It reflects how easy the data are understood.

Even though database is now accurate, complete, consistent and timely but low quality is due to poor believability & interpretability.

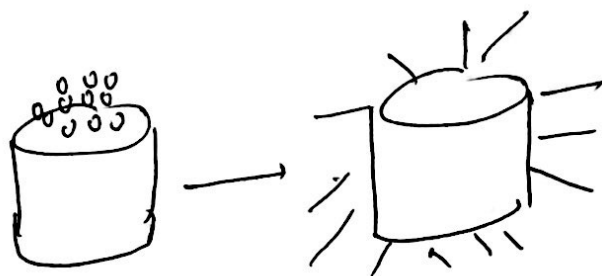
Unit-2Data preprocessing

Data we wish to analyse by data Mining technique are incomplete (lacking attribute value or certain attributes) of interest, noisy (containing error) & inconsistent. Data preprocessing technique improve the quality of data thereby help to improve accuracy & efficiency of Mining process.

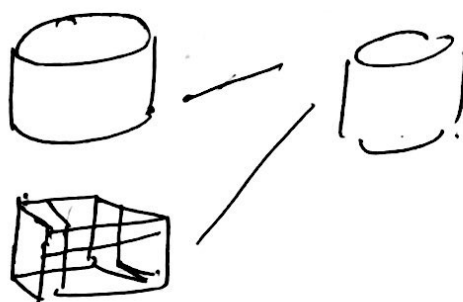
Process:

Data preprocessing techniques steps are

Data cleaning:-



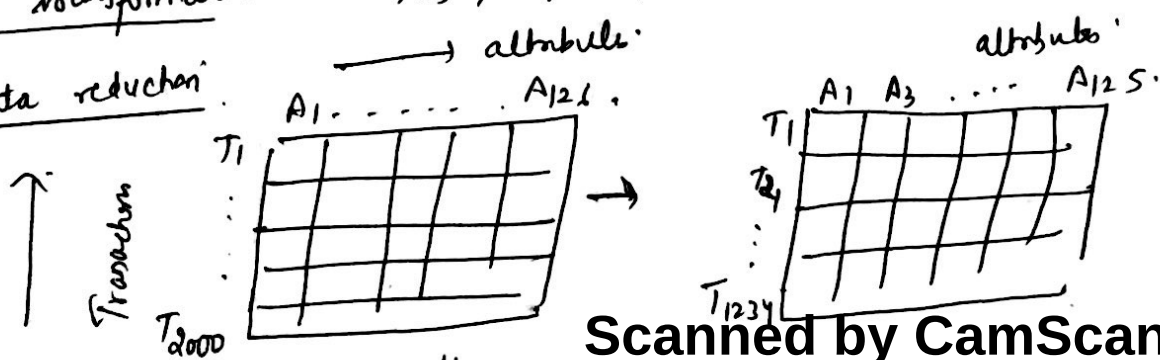
① Data Integration



3) Data transformation:

2, 63, 100, 51, 48 $\rightarrow$ 0.02, 0.063, 1.00, 0.51,

④ Data reduction



2.5

Data cleaning

cleaning is a process used to identify, incomplete or
irrelevant data & then improve quality by detecting errors.

Phases of data cleaning:-

- ① Analysis: To identify errors & Inconsistencies in data base.
- ② Defining transformation and Mapping Rules: In this we will find various problems that translate a list of activities as a result of analysis phase.
- ③ Verification:- The correctness & effectiveness should be tested & evaluated.
- ④ Transformation
- ⑤ Backflow of cleaned data:- After errors are removed, then clean data replace dirty data in original source.

a) Missing values :- Missing value prob occur when many tuples may have no record for several attributes.
How to fill the missing value for attribute?

(i) ignore the tuple: This is usually done when clean label is missing

(ii) Fill the missing value manually.

(iii) Use the attribute mean to fill missing value.

(iv) Use the most probable value to fill missing value.

(v) Use the attribute mean or median for all sample.

2.7.
cong: outliers may be detected by clustering. where minor values are organised into groups or clusters. The values which fall outside of set of clusters may be consider outliers.

Data Integration

Data Mining often require data integration - the merging of data from multiple data stores.

There are number of issues to consider during data integration like schema Integration & object matching.

How can equivalent Real world entities from multiple data sources to be matched up?

This is known as Entity Identification Problem.

for eg:- how can data analyst or computer is sure that customer-id in one database & customer-number in another refers to same entity.

2) Redundancy:- It is another important issue. An Attribute may be redundant if it is derived from another attribute or set of attributes.

3) Detection & Resolution of Data Value Conflicts: Third issue in data Integration is detection & Resolution of data value conflicts. For eg:- for some Real world entity, attribute values from different source may differ. This may be due to difference in representation, in scaling or encoding.

Data Transformation

data transformation, data are transformed into forms appropriate for Mining. Transformation, it involves following

- 1) Smoothing :- It tries to remove noise from data.
2. Aggregation :- where summary or aggregation operations are applied to data. For eg:- sales data may be aggregated so as to compute monthly & Annual total amounts.
- 3) Generalisation :- Discretization :- where raw values of numerical attribute (eg: age) are replaced by interval labels (eg: 0-10, 11-20 etc). (Age - young, middle age, senior).
- 4) Attribute construction :- where new attributes are constructed & added from given set of attributes to help Mining process. This simplify the original data & make the mining more efficient.
- 5) Normalisation :- Data Normalisation Means transforming all variables in the data to specific Range. It involve transforming data to fall within a smaller or common Range. $[-1, 1]$.

Technique used for Normalisation.

- ① Min-max Normalization
- ②. Z-score Normalisation.

$$= \frac{V - \min A}{\max A - \min A} (\text{new} \max A - \text{new} \min A) + \text{new} \min A.$$

suppose the minimum & maximum value for attribute income are Rs 12,000 and Rs 98,000. We would like to map income to range $[0, 1.0]$. By min-max normalisation, a value of Rs 73,600 for income is transformed to.

$$\frac{73,600 - 12,000}{98,000 - 12,000} (1.0 - 0) + 0 = 0.716.$$

②. Z-score normalisation (Zero Mean Normalisation) :-
This method is useful when actual minimum & maximum of attribute A are unknown.

Ques Suppose mean & standard deviation of values for attribute income are Rs 54,000 & Rs 16,000. With Z-score normalisation, a value of Rs 73,600 for income is transformed to.

$$V' = \frac{V - \bar{A}}{\sigma_A}$$

$$\frac{73,600 - 54,000}{16,000} = 1.225.$$

③ Discretisation :- It can be used to reduce number of values for given continuous attribute by dividing range of attribute into interval.

Data Reduction.

2.10

Data Reduction technique can be applied to obtain a reduced representation of data set that is much smaller in volume yet closely maintain integrity of original data.

Strategies for data reduction

- (1) Data cube aggregation :- where aggregation operations are applied to data in the construction of data cube
- (2) Dimension Reduction :- where irrelevant, weakly relevant may be detected & removed i.e. remove unimportant attribute.
- (3) Data Compression :- where Encoding mechanism has been used to reduce data size. (Huffman Encoding & run length Encoding).
- (4) Numerosity Reduction :- where data are replaced or estimated by alternative, smaller data representation such as parametric model. (which need store only the model parameters instead of actual data. or non parametric method such as clustering, sampling & use of histogram.
- (5) Discretization & Concept Hierarchy Generation →

Data Reduction

2.91

Reduction technique can be applied to obtain a Reduced representation of data set that is much smaller in volume, yet closely maintain integrity of original data.

Strategies for Data Reduction

(1) Dimensionality Reduction:- Dimensionality reduction reduce the data set size by removing such attributes from it. Typical method of attribute subset selection are applied. Attribute subset selection is a process in which subset of H attributes out of N attributes is chosen.

(2) Reduce the number of attributes appearing in the discovered pattern, helping to make pattern easier to understand. It Reduce data size by removing irrelevant or redundant attributes.

(3) Stepwise Forward selection:- Procedure starts with empty set of attributes. The best of the original attributes is determined and added to set.

Initial attribute set:
 $\{A_1, A_2, A_3, A_4, A_5, A_6\}$.

Initial reduced set:

$\{ \}$
 $\Rightarrow \{A_4\}$
 $\Rightarrow \{A_1, A_4\}$
 $\Rightarrow$ Reduced attribute set
 $\{A_1, A_4, A_6\}$.

backward.

Initial attribute set
 $\{A_1, A_2, A_3, A_4, A_5, A_6\}$.

$\Rightarrow \{A_1, A_3, A_4, A_5, A_6\}$

$\Rightarrow \{A_1, A_4, A_5, A_6\}$

Reduced attribute set:
 $\{A_1, A_4, A_6\}$.

2) stepwise backward elimination: This procedure starts with full set of attributes. At each step, remove the worst attribute remaining in the set.

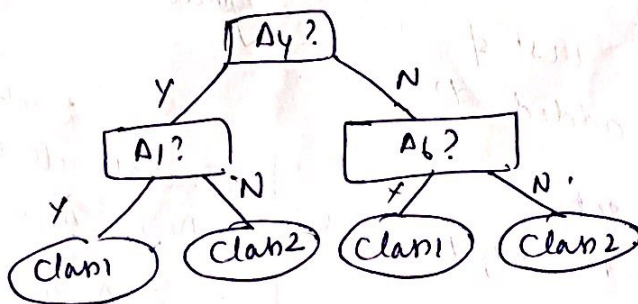
3) Combination of forward selection and backward selection.

The stepwise forward selection and backward elimination methods can be combined so that procedure selects best attribute & removes worst from among the remaining attributes.

4) Decision tree induction: When decision tree induction is used for attribute subset selection, a tree is constructed from given data. All attributes that do not appear in tree are assumed to be irrelevant.

Initial attribute set

$\{A_1, A_2, A_3, A_4, A_5, A_6\}$.



Reduced attributes are
 $\{A_1, A_4, A_6\}$.

data Reduction:- This technique can be used to reduce data volume by choosing smaller forms of data Representations.

Parametric Method:-

(1) Assume data fits in some Model,

In linear Model, data are modeled to fit a straight line.

for eg:- A Random variable y can be modeled as linear function of another Random variable x .

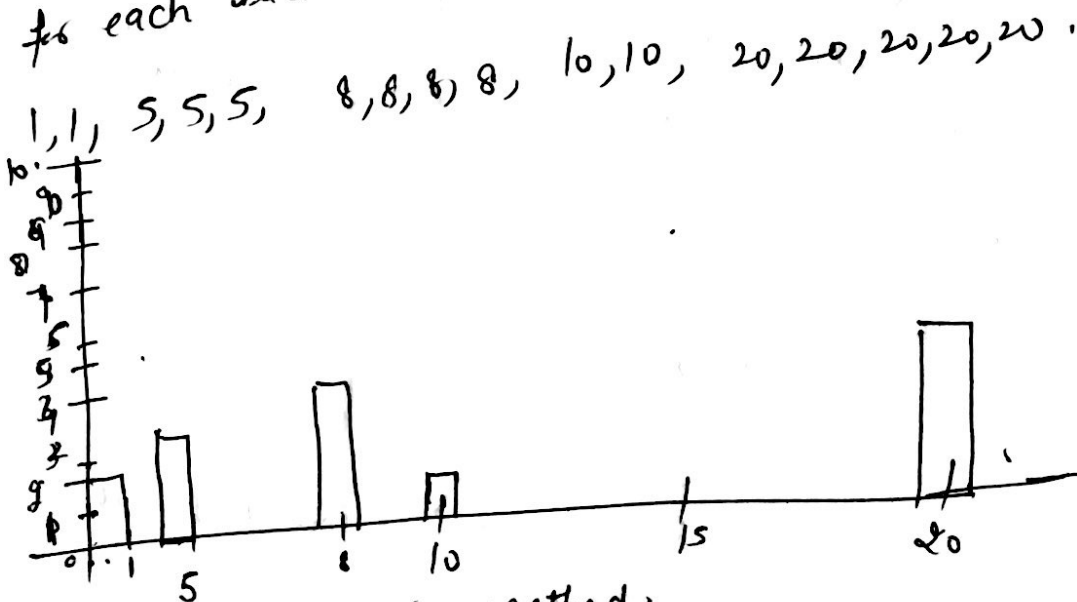
$$y = wx + b.$$

($y = y$ axis variable
 $x = x$ axis variable
 $w =$ Slope of line
 $b =$ Intercept)

Non Parametric Method:- In this there is no assumption of Models, It is used for storing reduced representation of data include histogram.

Major families are histogram, clustering

(i) Histogram: (i) Popular data reduction technique.
 (ii) divide data into buckets & store average (sum) for each bucket.



Histogram use binning method.

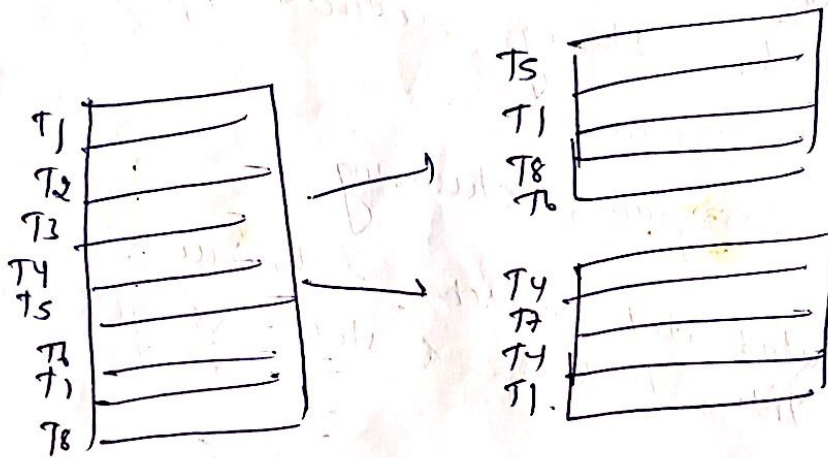
① Clustering :- clustering technique consider data as objects. They partition the objects into groups, so that objects within a cluster are similar to one another and dissimilar to objects in other clusters.

→ In data Reduction, cluster representation of data are used to replace

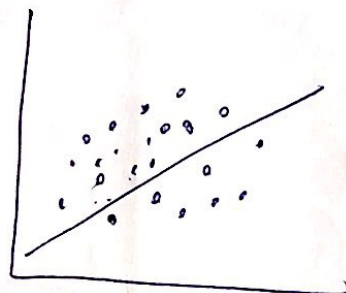
Sampling

Sampling can be used as a data Reduction technique because it allows large data set to be represented as by much smaller random data sample, or subsets.

eg:-



(The term linear refers to the fact that we are fitting a line. Model refers to the equation that summarizes the line that we fit.)



Linear method enable you to determine relationship between continuous process output (Y) & one factor (X).

Discretization and Concept Hierarchy Generation 2.15.

Discretization technique can be used to Reduce the number of values for a given continuous attribute by dividing range of attributes into intervals. Interval labels can be used to replace actual data values.

Concept Hierarchy

It reduce the data by collecting and replacing low level concept (such as numeric values for attribute age) by higher level concept (such as young, middle age or senior).

Discretization and Concept Hierarchy Generator for Numeric Data.

- (1) Binning :- Binning or discretization is the process of transforming numerical variable into categorical counterpart. It is top down splitting technique based on specified number of bins.
- (2) Histogram Analysis.. Like Binning, histogram partitions the value for an attribute A, into disjoint ranges called bucket.

How are buckets determined, and attribute values partitioned, there are several Rules.

- ① Equal Frequency Partitioning
- ② Equal width Partitioning

Suppose a group of 12 sales price records has been sorted as follow.

5, 10, 11, 13, 15, 35, 50, 55, 72, 92, 204, 215.

Partition them into three bins

- (i) equal frequency partitioning (iii) clustering.
 (ii) Equal width partitioning

(i) Equal frequency Partitioning

bin 1	5, 10, 11, 13.
bin 2	15, 35, 50, 55
bin 3	72, 92, 204, 215.

(ii) Equal width Partitioning:-

width of interval is $(215 - 5) / 3 = 70$.
 $(\text{max} - \text{min}) / N$.

width of equal width partitioning is

bin 1	5, 10, 11, 13, 15, 35, 50, 55, 72
bin 2	92
bin 3	204, 215.

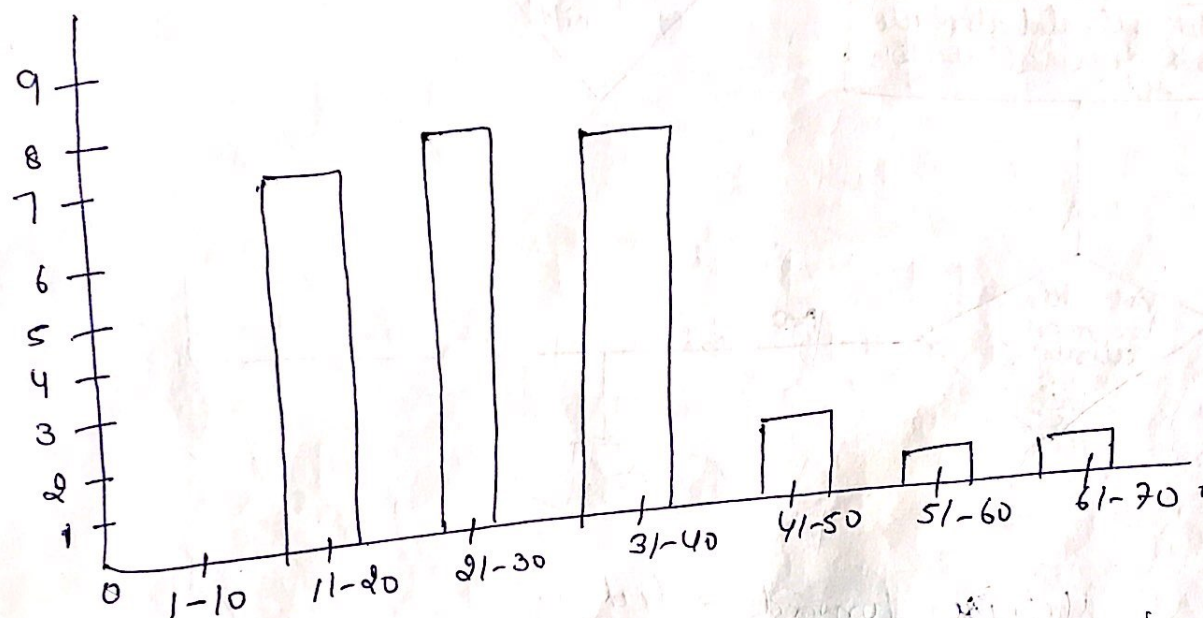
(iii) clustering → Partitioning the data into 2 biggest gap.

bin 1	5, 10, 11, 13, 15.
bin 2	35, 50, 55, 72, 92
bin 3	204, 215.

that the data for analysis include the age. The age values for the data tuples are in increasing order.

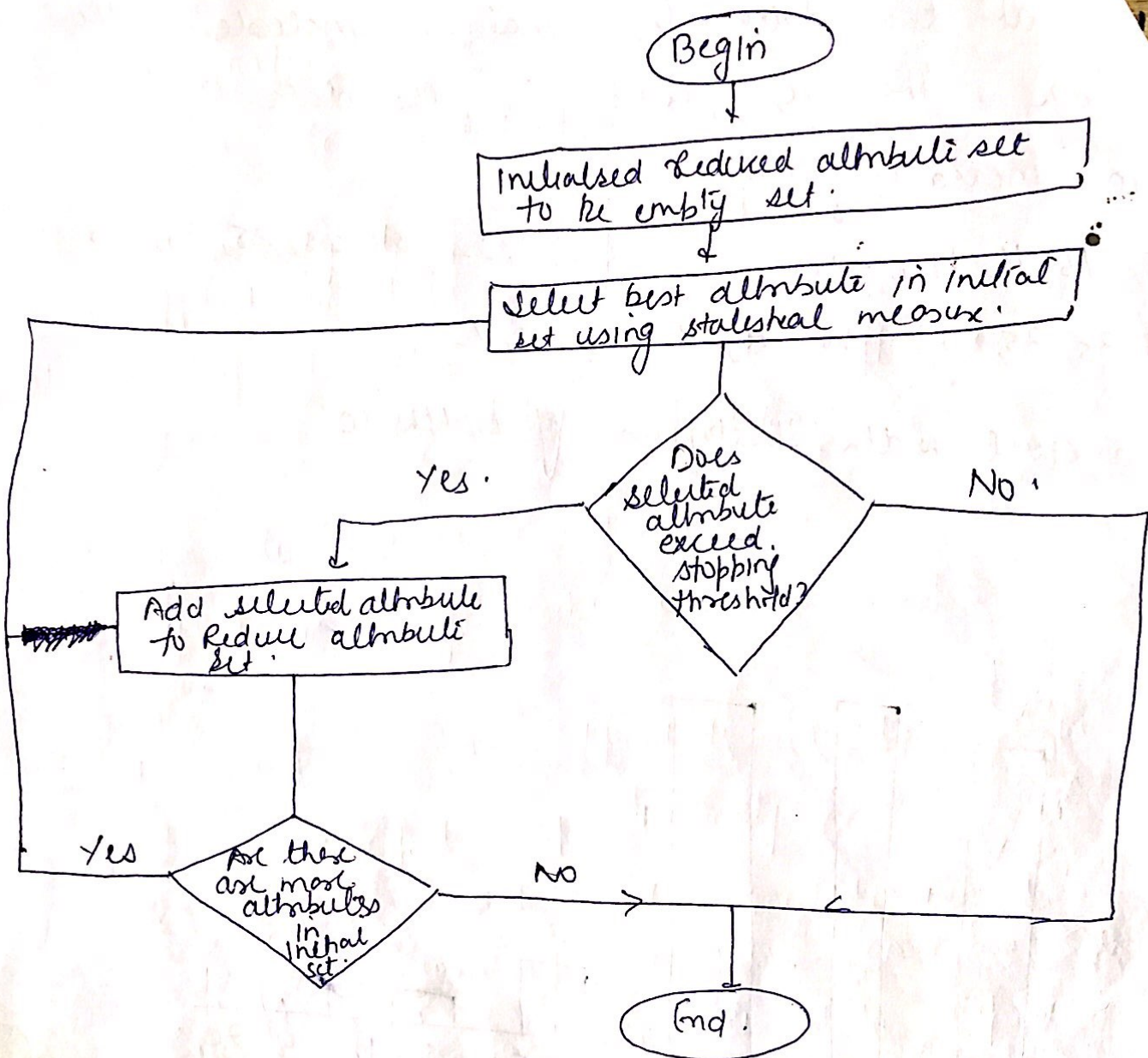
13, 15, 16, 16, 19, 20, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30
33, 33, 35, 35, 35, 36, 40, 45, 46, 52, 70.

Plot an equal width histogram of width 10.



Ques: Use a flow chart to summarise following procedure.
for attribute selection.

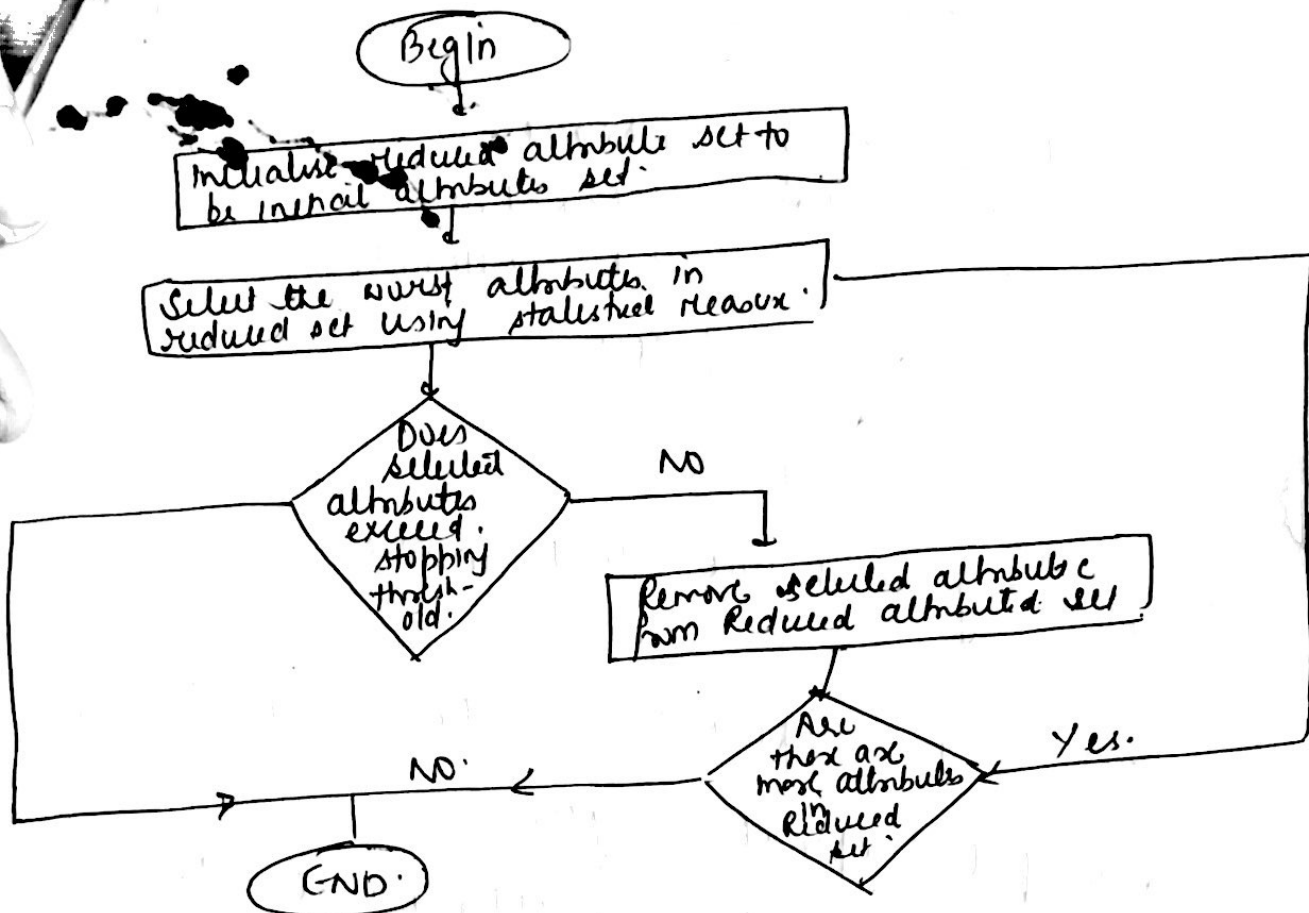
- (1) stepwise forward selection
- (2) stepwise backward selection.



stepwise forward selection.

Stepwise Backward Elimination

2/19



Data Compression

Data compression technique reduce dimensionality of the data. Compression technique are broadly classified as lossless & lossy. When the original data can be reconstructed entirely from condensed form are referred as lossless. Where only an approximate of original data can be constructed are term as lossy.

Lossless Compression → Technique such as run length encoding permit basic and minimal reduction of data size.

Lossy technique → Technique such as Principal component analysis, wavelet transform can be used.

Wavelet Transform

DWT (Discrete wavelet transform)

It is linear signal processing technique. When applied to data vector D , transform it into different vectors D' of wavelet coefficients. DWT achieve, better lossy compression.

DFT (Discrete Fourier Transform)

It is signal processing technique that involve sines & cosines.

Wavelet transform can be applied to multidimensional data.

Principal Component Analysis

PCA is technique used to reduce multidimensional data set to lower dimensions for analysis. PCA is a method of identifying patterns in data and data representation of format that highlight the similarity & dissimilarity of data. It reduce dimension number of dimension without much loss of information.

Mining Frequent Patterns, Associations and Correlations

Frequent Patterns are patterns (eg: itemset, subsequences or substructures) that appear frequently in data set. For eg:- set of items such as milk and bread. that appear frequently together in transaction data set is a frequent itemset.

A subsequence such as buying first PC, then digital camera and then memory card, if it occurs frequently in a shopping history database called sequential pattern.

Mining Association rule Mining

Association Rule Mining finds interesting association or correlation / relationship among large set of data items. Association rules show attribute value conditions that occur frequently together in given data set.

The discovery of interesting correlation relationship among huge amounts of business transaction records can help in many business decision making processes such as catalog design, cross marketing & customer shopping behaviour analysis.

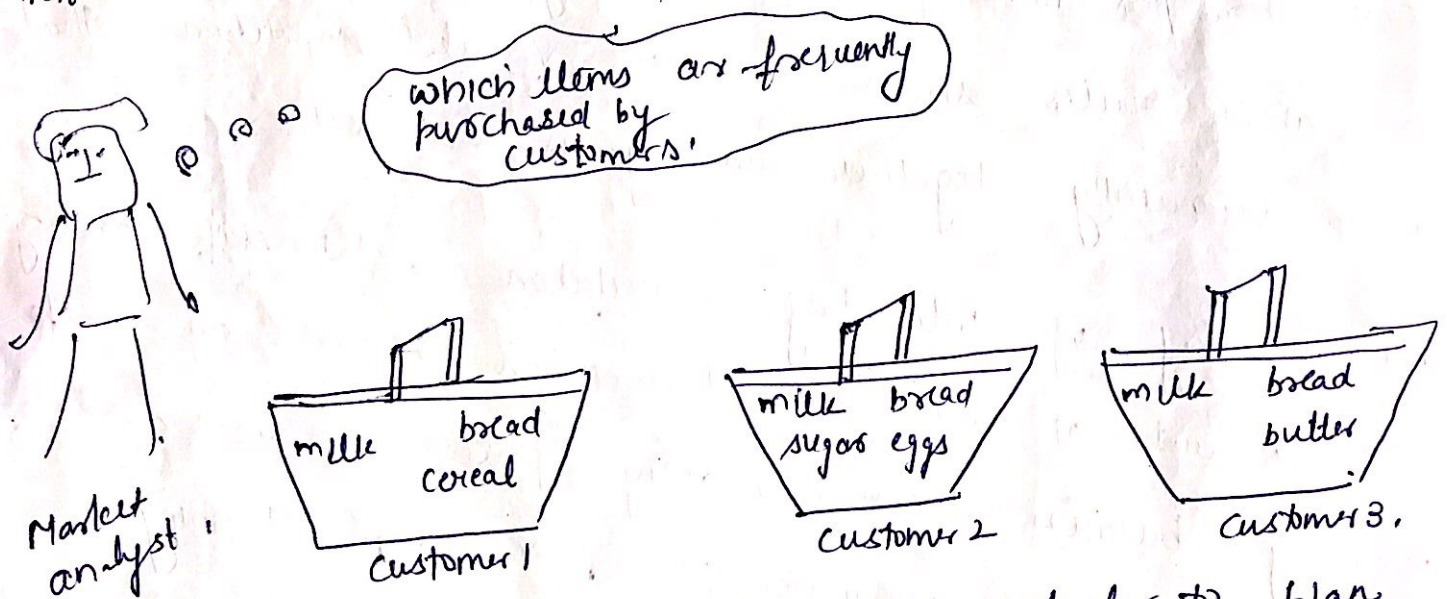
Market Basket Analysis :- An Example.

Market Basket Analysis (Association Analysis) is a mathematical modelling technique based upon the theory that if you have buy certain group of items, you are likely to buy another group of items.

This process analyses customer's buying habits by finding association between different items that customer place in their shopping baskets.

For eg:- If customers are buying milk, how likely are they to also buy bread.

→ The set of items a customer buy is referred as an itemset & market basket analysis seek to find relationships between purchases.



Market basket Analysis can also help retailers to plan which items to put on sale at reduced prices.

Market Basket Analysis

Mathematical Modelling technique based upon the theory if you buy a certain group of items, you are likely to buy another group of items.

The set of items a customer buy called itemset.

Market Basket Analysis seek to find out relationship between purchases. It is used to analyse customer purchasing behaviour and help in increasing the sales.

Market Basket Analysis can also help retailer to plan which item to put on sale at reduced price.

If we think universe as the set of items available at store. Each item has a boolean variable representing presence & absence of that item. Each basket is represented by a Boolean vector of values assigned to this variable.

The Boolean vector can be analysed for buying pattern which recent items that are frequent associated or purchased together. These pattern can be represented in form of association rules.

Association rules are in the form of if X then Y .

Consider a Market with collection of huge customer transaction. X and Y are set of items & rule means the customer

Association Rules.

3.2

Association rules are of the form if X then Y .

eg:- 80% of people. The information that customers who purchase computers also tend to buy antivirus software at the same time is represented in following Association Rule.

Computer $\Rightarrow$ antivirus software.

Areas in which association rules have been used, include:

1. Credit card transactions.
2. Supermarket purchases.
3. Banking services.
4. Insurance patterns.
5. Medical patient histories.

An association rule is $X \rightarrow Y$ where X is called antecedent and Y is consequent. X and Y are the sets of items and the rule means that customer who buy X are likely to buy Y with probability $\% c$ where c is called confidence.

For eg: 80% of people who buy cigarettes also buy matches.

If (Bread) Then (butter).

These are two types of Association Rules

- 1) Support level
- (2) Confidence level.

Support: - A - measurements of items that appear together (purchased together).

$$\text{Support } (A \rightarrow B) = P(A \text{ and } B),$$

Support indicate how frequently the items appears in database.

$$\text{Support } (A \Rightarrow B) = \frac{P(A \wedge B) \cdot \text{Number of tuples containing both A and B}}{\text{Total no of tuples}}$$

Suppose. U and T are the number of elements in U & T.

Confidence: - It is measurement of tendency to buy item after another one. $\text{Confidence } (A \rightarrow B)$
 $= \frac{\text{No of tuples containing both A and B}}{\text{No of tuple Contain A}}$

For eg:- Computer $\rightarrow$ antivirus - software [support = 2%, confidence = 60%].

A support of 2% for Association Rule means that 2% of all the transaction under analysis show that Computer & antivirus software are purchased together.

Confidence of 60% means that 60% of customer who purchased a computer also bought the software.

Classification of Association Rules.

3.3.

Based on type of values handled in the Rule.

If a rule concerns association between presence or absence of items, it is a Boolean association rule. If a rule describes association between quantitative items or attributes then it is quantitative association rule. Example of -

② Based on the dimension of data involved in Rule
If the items or attribute in association rule each reference only one dimension then it is single dimensional Association Rule.

buy (x, "Computers") $\rightarrow$ buys (x, "HP ~~Printer~~ Printer").
artificial software.

If two or more dimensions

age (x, "30...39") $\wedge$ Income (x, "42...48k") ?

buys (x, "high resolution TV").

Rules buys (x, "computers") buys (x, "HP Printer") are Boolean association Rule.

Quantitative means,

age (x, "30...39") $\wedge$ Income (x, "42k...48k") $\rightarrow$

buys (x, "high resolution TV").

③ Based on level of Abstraction

Some methods for abstraction. Rule mining can find at different level of abstraction

age (x, "30-34") $\Rightarrow$ buys (x, "laptop computers")

age (x, "30-34") $\Rightarrow$ buys (x, "computer").

Rule mine
The set of items
and a small database
presence and 0 abso
An example Rule
of mine
bread
In

Examples to understand support and Confidence

use small example from supermarket domain.
The set of item set is $I = \{ \text{milk, bread, butter, juice} \}$
and a small database containing the items (1 code presence and 0 absence of an item in transaction).

An example Rule for supermarket could be

$\{ \text{milk, bread} \} \Rightarrow \{ \text{butter} \}$ meaning that if milk & bread is brought, customers also buy butter

In practical, a rule need a support of several hundred transactions. & data set contain often contain million of transactions.

$$\{ \text{milk, bread} \} \Rightarrow \{ \text{butter} \}$$

$$\{ x \} \Rightarrow \{ y \}$$

Transaction Id	milk	bread	butter	Juice
1	1	1	0	0
2	0	1	1	0
3	0	0	0	1
4	1	1	1	0
5	0	1	0	0
6	1	0	0	0
7	0	1	1	1
8	1	1	1	1
9	0	1	0	1
10	1	1	0	0

sup (x)

Support: $\text{supp}(x)$ of an itemset, x
no of transactions which contain itemset (x)

total no of transaction.

Itemset (milk, bread, butter) has support of $= \frac{9}{10} = \frac{9}{5} = 0.9$
since it occur in 90% of transaction.

Confidence of Rule is defined as.

$$\text{conf}(X \rightarrow Y) = \frac{\text{support}(X \cup Y)}{\text{support}(X)}$$

$$\frac{\text{support}(\text{milk, bread, butter})}{\text{support}(\text{milk, bread})}$$

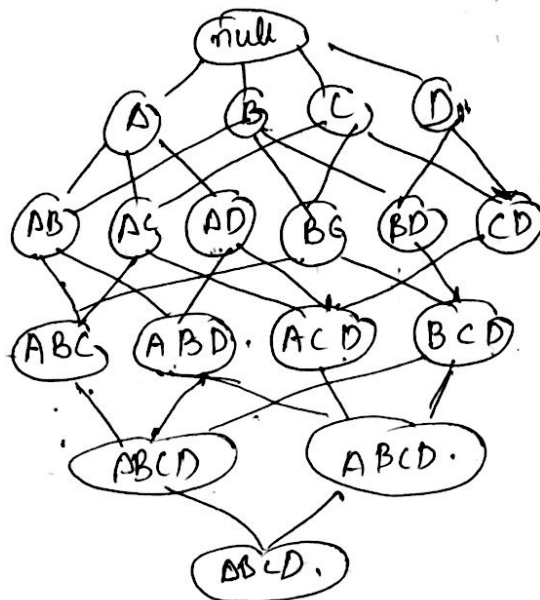
$$\text{Support}(\text{milk, bread, butter}) = \frac{2}{4} = 0.5 \Rightarrow 50\%$$

$$\text{Support}(\text{milk, bread}) = \frac{4}{10} = 0.4 \Rightarrow 40\%$$

$$\text{Confidence} = \frac{0.5}{0.4} = \frac{5}{4} = 1.25$$

$\Rightarrow 50\%$

Frequent itemset Generation



Algorithm for Mining single Dimensional Boolean Association rules from Transactional databases.

Apriori is a classic algorithm for learning association rules. The name of the algorithm is based on the fact that the algorithm uses prior knowledge of frequent itemset properties.

An Iterative approach known as level wise search is used. In this algorithm, k itemsets are used to explore $(k+1)$ elements of frequent itemset. To improve the efficiency of level wise generation of frequent itemset, apriori property is used.

- (1) Apriori property state that all non empty subset of frequent itemset must be frequent.
- (2) The property used in this algorithm uses two step process. It consist of join step and prune step.

First the set of frequent 1-itemset is found. This set is denoted by L_1 , L_1 is used to find L_2 , the frequent 2-itemset which is used to find L_3 . The finding of each L_k require full scan of database.

How Apriori ~~algorithm~~ ^{property} used in algorithm. To understand this. Let us look at how L_{k-1} is used to find L_k for $k \geq 2$. A two step process is followed consist of join and prune action.

1) Join step: To find L_k , a set of candidate k -elements generated by joining L_{k-1} with itself. This set of candidates is denoted by C_k . Let L_1 and L_2 be element in L_{k-1} .

2) Prune step: C_k is subset of L_k , i.e. its members may or may not be frequent but all of its frequent k -element are included in C_k . A scan of database to determine the count of each candidate in C_k would result in the determination of L_k .

... of Apriori, based on transaction database, D.

TID.	list of items.
T ₁₀₀	I ₁ , I ₂ , I ₅
T ₂₀₀	I ₂ , I ₄
T ₃₀₀	I ₂ , I ₃
T ₄₀₀	I ₁ , I ₂ , I ₄
T ₅₀₀	I ₁ , I ₃
T ₆₀₀	I ₂ , I ₃
T ₇₀₀	I ₁ , I ₃
T ₈₀₀	I ₁ , I ₂ , I ₃ , I ₅
T ₉₀₀	I ₁ , I ₂ , I ₃

There are nine Transactions in database i.e. $|D| = 9$.
 The algorithm simply scans all of the transactions in order to
 count the number of occurrence of each item.
 In first iteration of algorithm, each item is a member of
 set of candidate 1-itemsets C_1 .

scan D. for
 count of each
 candidate.

Item set	Support count
{I ₁ }	6
{I ₂ }	7
{I ₃ }	6
{I ₄ }	2
{I ₅ }	2

Compare
 candidate
 support
 count with
 minimum
 support
 count.

L_1

Item set	Support count
{I ₁ }	6
{I ₂ }	7
{I ₃ }	6
{I ₄ }	2
{I ₅ }	2

Suppose the minimum support count required is 2.
 Corresponding relative support is $\frac{2}{9} = 22\%$.
 minimum support count = 22%.

The set of frequent 1-itemsets is L_1 .

all C_2 candidates from L_1 →

C_2

Itemset
$\{I_1, I_2\}$
$\{I_1, I_3\}$
$\{I_1, I_4\}$
$\{I_1, I_5\}$
$\{I_2, I_3\}$
$\{I_2, I_4\}$
$\{I_2, I_5\}$
$\{I_3, I_4\}$
$\{I_3, I_5\}$
$\{I_4, I_5\}$

Scan D for count of each candidate →

C_2

Itemset	Supp-count
$\{I_1, I_2\}$	4
$\{I_1, I_3\}$	4
$\{I_1, I_4\}$	1
$\{I_1, I_5\}$	2
$\{I_2, I_3\}$	4
$\{I_2, I_4\}$	2
$\{I_2, I_5\}$	2
$\{I_3, I_4\}$	0
$\{I_3, I_5\}$	1
$\{I_4, I_5\}$	0

3.

Compare candidate support count with minimum support.

L_2

Itemset	Sub-count
$\{I_1, I_2\}$	4
$\{I_1, I_3\}$	4
$\{I_1, I_5\}$	2
$\{I_2, I_3\}$	4
$\{I_2, I_4\}$	2
$\{I_2, I_5\}$	2

L_2 represent set of frequent 2-itemset.

Generate C_3 candidate from L_2 →

C_3

Itemset
$\{I_1, I_2, I_3\}$
$\{I_1, I_2, I_5\}$

Scan D for count of each candidate →

C3

Item set	Sub Count
$\{I_1, I_2, I_3\}$	2
$\{I_1, I_2, I_5\}$	2

Compare candidate support with minimum support count

L3

4.

Itemset	Sub Count
$\{I_1, I_2, I_3\}$	2
$\{I_1, I_2, I_5\}$	2

Two operations

① Join operation : $L_2 \bowtie L_2$ (Self join)

$\{\{I_1, I_2\} \{I_1, I_3\}, \{I_1, I_5\}, \{I_2, I_3\}, \{I_2, I_4\}, \{I_2, I_5\}\} \bowtie$
 $\{\{I_1, I_2\}, \{I_1, I_3\}, \{I_1, I_5\}, \{I_2, I_3\}, \{I_2, I_4\}, \{I_2, I_5\}\}$

$\Rightarrow \{ \{I_1, I_2, I_3\}, \{I_1, I_2, I_5\}, \{I_1, I_3, I_5\}, \{I_2, I_3, I_4\},$
 $\{I_2, I_3, I_5\}, \{I_2, I_4, I_5\} \}$

(b) Apriori Property :- All non empty sets of frequent itemsets must also be frequent.

(1) 2 item set subset of $\{I_1, I_2, I_3\}$ are $\{I_1, I_2\}, \{I_1, I_3\}$ and $\{I_2, I_3\}$. $\Rightarrow$ All 2 item subsets of $\{I_1, I_2, I_3\}$ they will keep in C3.

(2) The 2 item subset of $\{I_1, I_2, I_5\}$ are $\{I_1, I_2\}, \{I_1, I_5\}$ and $\{I_2, I_5\}$. All 2 item subsets of $\{I_1, I_2, I_5\}$ they will keep in C3.

Four letter candidates cannot be possibly frequent.